costing_DESIGN_REVIEW_LOG

Design review log (2026-11-02): defects found by the in-depth review and what was done

Method: automated checks (`utility10/pcb/review.py` part identity, stock, capacitor voltage, LED current, decoupling, pull-ups; `utility10/pcb/pincheck.py` supply and ground pin names against the nets) plus a manual read of every diode, polarised part and transistor against the KiCad symbol and footprint pin conventions (diode pin 1 = cathode, pad 1 = cathode).

Defects found and fixed

#	Where	Defect	Effect if built	Fix
1	MITTI-NETRA D1, D2	Schottky diodes from solar and USB into the charger were fitted backwards (cathode on the source)	The TP4056 would never receive power from solar or USB	Nets swapped and part rotated 180 degrees; copper unchanged; schematic regenerated; checks re-run
2	KINETIC controller D5, BioTray D5	The "USB 5 V into +5V" diode was backwards (cathode on VBUS)	The 5 V buck output would back-feed the USB port, and USB could not power the logic	Same fix
3	Parts at LCSC	Out of stock: 100 nF 0603 (C14663), 10 nF 0603 (C57112), Everlight 0603 LEDs, WS2812B-Mini, 1x2 header; very low stock: 100 uF 35 V (85), 2x3 header (340)	Order would fail at the fab or stall	Replaced by in-stock equivalents with the same footprint (see `manuals/ORDERING_GUIDE.md`)
4	Green indicator LED	Everlight 518 nm green (Vf about 3.3 V) from a 3.3 V rail	Almost dark	Yellow-green KENTO KT-0603YG (Vf about 2 V)
5	Gas-Suraksha demo firmware	Drives the buzzer with plain on/off, which silences a passive magnetic buzzer	No sound	New board uses an ACTIVE buzzer (see its manual)

Checked and found correct

Flyback diodes (cathode to supply, anode to the driven node) on all buzzer, pump, fan and valve drivers; unidirectional TVS orientation (cathode to the positive rail); gate-clamp Zener orientation on the reverse-polarity PMOS stages; P-channel reverse-polarity stages (drain on the battery side); polarised capacitor orientation; LED orientation; NPN (B, E, C) and PMOS (G, S, D) pin order against the SOT-23 footprints; AP2112K pin order; supply and ground pin names of every IC against its symbol (0 errors on all 20 boards).

Open recommendations (not blocking)

- Passive magnetic buzzers (12 mm, 16 ohm) are driven at 4.2 to 4.5 V from a transistor: the estimated current is above the 30 mA rating for continuous use; fine for short beeps, add a 22 ohm series resistor if the buzzer gets warm.

- Order confidence for BQ25570 (stock about 390), the 12 mm passive buzzer (140) and PESD5V0S1BA (175) is limited for volume orders.
- Nothing has been built: every conclusion here comes from files and datasheet reading, not from measurement.